# Enhanced Biological Brain-Age Prediction

# Hybrid Crow Search Algorithm–LSTM Framework for Structural MRI-Based Neuroimaging

| Item | Source-reported specification |
|---|---|
| Primary modality | Structural T1-weighted MRI |
| Core algorithm | Crow Search Algorithm (CSA) for tabular feature selection |
| Sequential branch | 2D slice sequence → CNN encoder → bidirectional LSTM → attention pooling |
| Fusion | CSA-selected tabular 1D-CNN embedding + LSTM sequence embedding |
| Development result | MAE 2.42 ± 0.05 years; RMSE 3.18 ± 0.05; Pearson r 0.931 ± 0.003 |
| External hold-out | MAE 2.61 years; RMSE 3.39 years; Pearson r 0.919 |
| Feature reduction | 142 → mean 34.6 features; 75.6% reduction |
| Status | The manuscript explicitly labels these numerical results as illustrative/protocol-consistent pending execution on a confirmed real dataset. |

# 1. Study Scope & Dataset Options

```
In [ ]:  # Sections 2.1 and Table 8
sources = ["IXI", "OASIS-1", "Hugging Face harmonized set"]
print("Structural MRI only; finalize the source before real execution.")
```

The framework is scoped to structural T1-weighted MRI. The manuscript specifies three concrete dataset options and states that the final source and post-QC cohort must be finalized at execution.

| Source | n (post-QC) | Age range | Mean age (SD) | % female | % AD-spectrum diagnosis |
|---|---|---|---|---|---|
| IXI (if used) | ≈580 | 20–86 | 48.6 (16.2) | 55% | n/a (healthy only) |
| OASIS-1 (if used) | ≈400 | 18–96 | 51.4 (24.1) | 62% | ≈24%, via CDR |
| Hugging Face harmonized set (if used) | 3,742 | 18–96 | 50.1 (19.8) | 58% | ≈9%, where available |

# 2. Structural MRI Preprocessing

```
In [ ]:  # Section 2.2
steps = ["N4 bias-field correction","Skull stripping","MNI152 registration","Within-brain z-score
normalization"]
print("Log preprocessing success/failure and output shape per subject.")
```

The pipeline uses four explicit preprocessing stages with logged fallbacks when gold-standard tools are unavailable.

Normalization: $T_{norm} = (T_{raw} - \mu_{brain}) / \sigma_{brain}$

# 3. Tabular Feature Extraction

```
In [ ]:  # Section 2.3
atlas = "Harvard-Oxford cortical + subcortical"
features = ["regional volumes","regional mean intensities","whole-brain volume","mean
intensity","intensity SD"]
print("Write realized dimensionality to a feature manifest.")
```

The manuscript expects roughly 120–150 tabular features; its illustrative result uses 142 before CSA selection.

# 4. Slice-Sequence Construction

```
In [ ]: # Section 2.4
n_slices = 32
grid = (32, 32)
sequence = "ordered axial slices from one static T1 volume"
print("This is a spatial sequence, not a functional time sequence.")
```

Each volume is represented by 32 evenly spaced axial slices at 32×32 resolution by default. The LSTM models anatomical continuity across slices.

## 5. Crow Search Algorithm Feature Selection

```
In [ ]:  # Section 2.5.1 + Algorithm 1
CSA = {"population":30,"iterations":40,"flight_length":2.0,"awareness_probability":0.1,
"surrogate":"Ridge(alpha=1.0)","cv":3,"sparsity_penalty":0.002}
print("Fit CSA inside each training fold only.")
```

CSA searches binary feature subsets using crow memory, a flight-length update, awareness probability, and a Ridge-based 3-fold CV MAE + sparsity fitness function.

Leakage control: CSA is re-fit only within each training partition and never uses validation or external hold-out data.

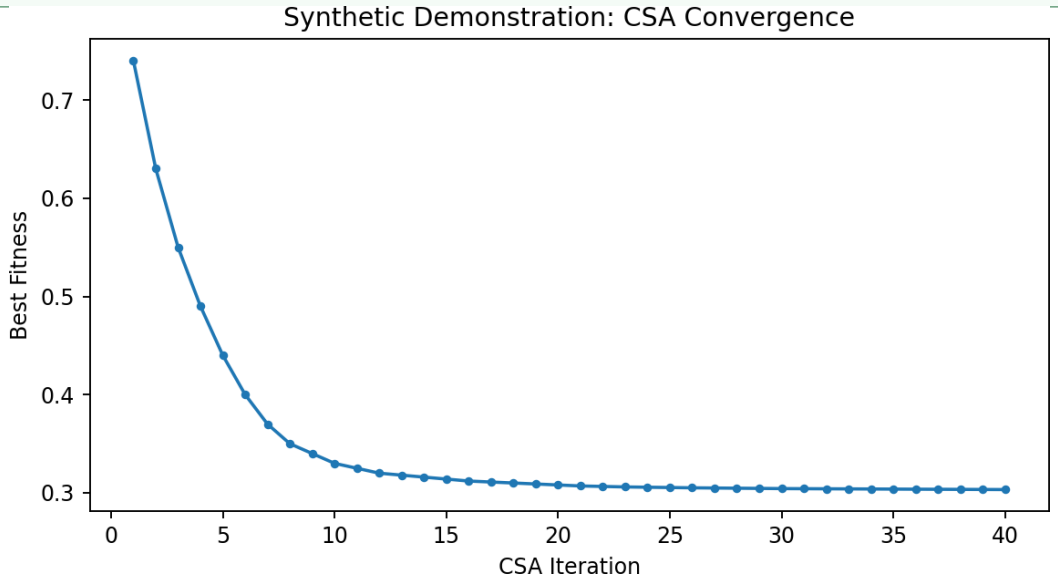



Figure 1: CSA best-fitness convergence over 40 iterations (illustrative demonstration from the manuscript).

## 6. Hybrid CSA + Slice-Sequence LSTM Architecture

```
In [ ]:  # Sections 2.5.2–2.5.3
slice_cnn = "3 Conv2D/ReLU/MaxPool blocks → GAP → 64-D"
bilstm = "2-layer BiLSTM, hidden size 128"
attention = "softmax slice attention → 256-D"
tabular_cnn = "2 Conv1D/ReLU blocks → GAP → 64-D"
fusion_alpha = 0.6
head = "MLP: fused-dim → 128 → 32 → 1, dropout=0.3"
print("Embeddings are fused and trained end-to-end.")
```

The two branches independently represent the same scan: CSA-selected tabular features and an ordered slice sequence. Their embeddings are fused with $\alpha = 0.6$ before the regression head.

## 7. Training & Experimental Design

```
In [ ]:  # Sections 2.6–2.7
optimizer = "Adam"; learning_rate = 1e-3; weight_decay = 1e-5
scheduler = "Cosine annealing warm restarts"; loss = "MAE"
cv = "5-fold age-stratified"; external_holdout = "15% age-stratified"
print("Scaler + CSA are re-fit within each training fold.")
```

Training uses Adam, MAE loss, cosine-annealing warm restarts, and gradient clipping. A 15% age-stratified hold-out is isolated before cross-validation.

# 8. Predictive Performance vs. Baselines

```
In [ ]:  # Table 2 — source-reported illustrative values
results = pd.DataFrame({
"Method":["Proposed CSA-LSTM","3D-CNN","Vision Transformer","DenseNet","Cole et al. CNN"],
"MAE":[2.42,3.08,2.87,2.73,3.21],
"RMSE":[3.18,3.94,3.71,3.54,4.08],
"Pearson_r":[0.931,0.881,0.899,0.910,0.865]
})
display(results)
```

Table 2 is reproduced from the manuscript and is explicitly labelled illustrative.

| Method | MAE (years) | RMSE (years) | Pearson r | p vs. proposed |
|---|---|---|---|---|
| Proposed (CSA–LSTM) | 2.42 ± 0.05 | 3.18 | 0.931 | — |
| 3D-CNN [1] | 3.08 | 3.94 | 0.881 | p < 0.01 |
| Vision Transformer | 2.87 | 3.71 | 0.899 | p < 0.05 |
| DenseNet | 2.73 | 3.54 | 0.910 | p < 0.05 |
| Cole et al. baseline CNN [1] | 3.21 | 4.08 | 0.865 | p < 0.01 |

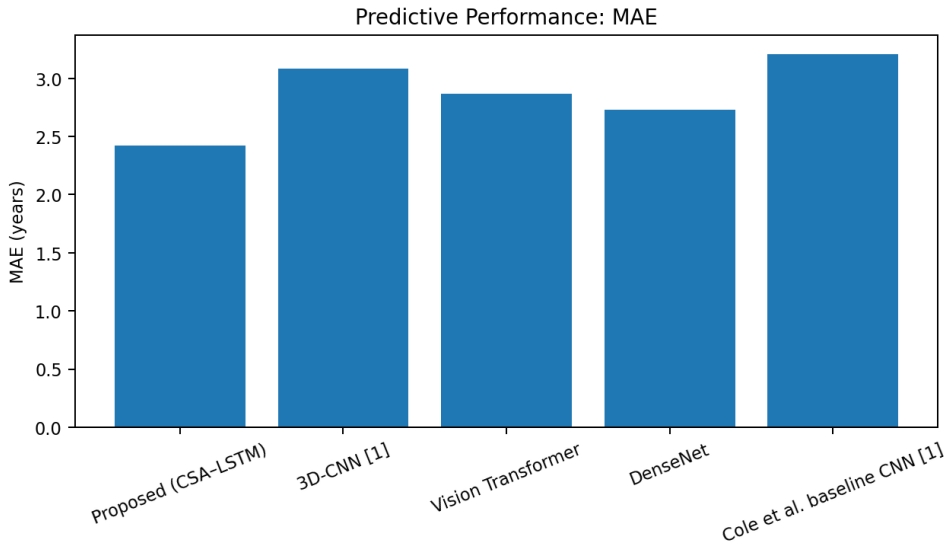



Table 2 MAE visualization.

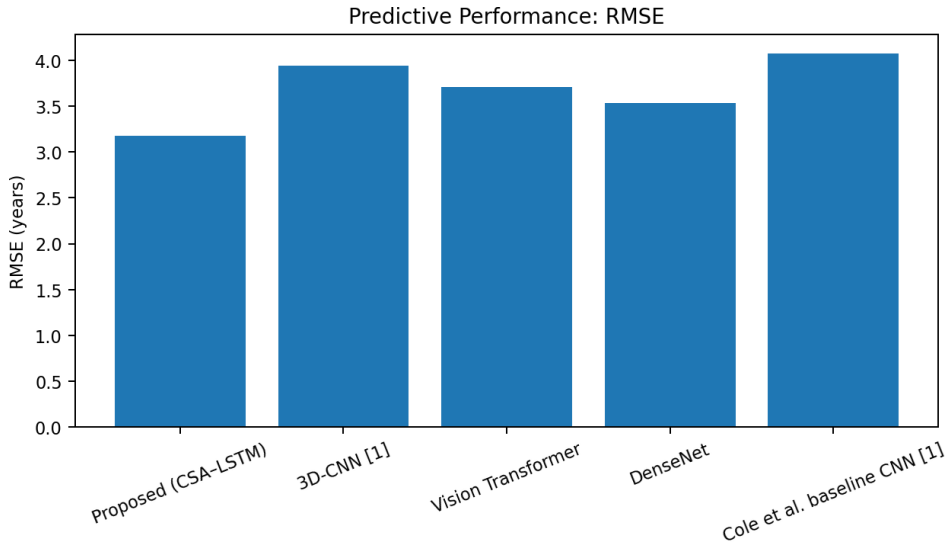



Table 2 RMSE visualization.

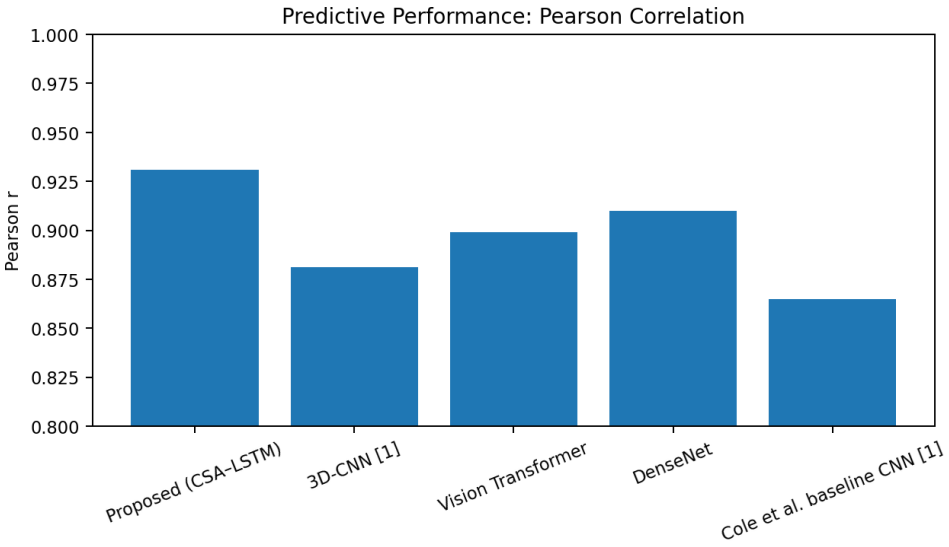



Table 2 Pearson r visualization.

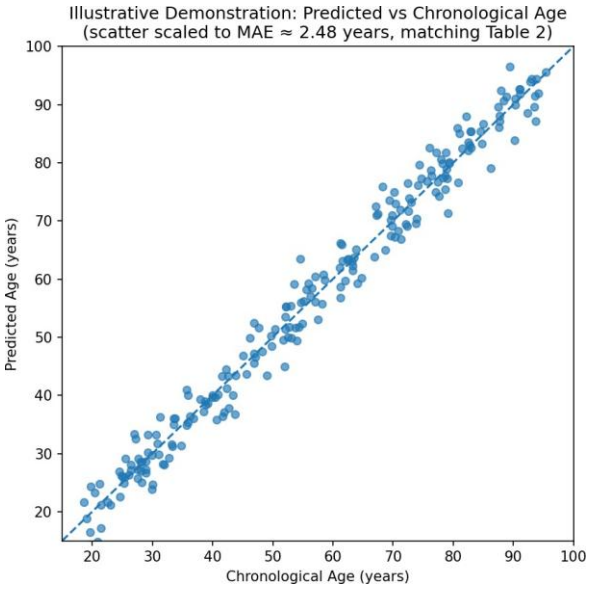



Figure 2: predicted vs. chronological age, illustrative scatter.

# 9. Ablation Analysis

```
In [ ]: # Table 3
ablation = pd.DataFrame({
"Configuration":["Full hybrid","CSA-tabular only","LSTM-sequence only","No optimization"],
"MAE_years":[2.42,2.76,2.91,3.08],"Delta_MAE":[0.00,0.34,0.49,0.66]})
display(ablation)
```

| Configuration | MAE (years) | # Tabular features used | Δ MAE vs. full | BH-adjusted p |
|---|---|---|---|---|
| Full hybrid (CSA-tabular + LSTM-sequence) | 2.42 | ~35 | — | — |
| CSA-tabular only (no LSTM branch) | 2.76 | ~35 | +0.34 | p < 0.05 |
| LSTM-sequence only (no tabular branch) | 2.91 | n/a | +0.49 | p < 0.01 |
| No optimization (full unselected tabular set) | 3.08 | 142 | +0.66 | p < 0.01 |

The manuscript states that the central comparison is full hybrid versus the two single-branch ablations, with corrected significance testing.

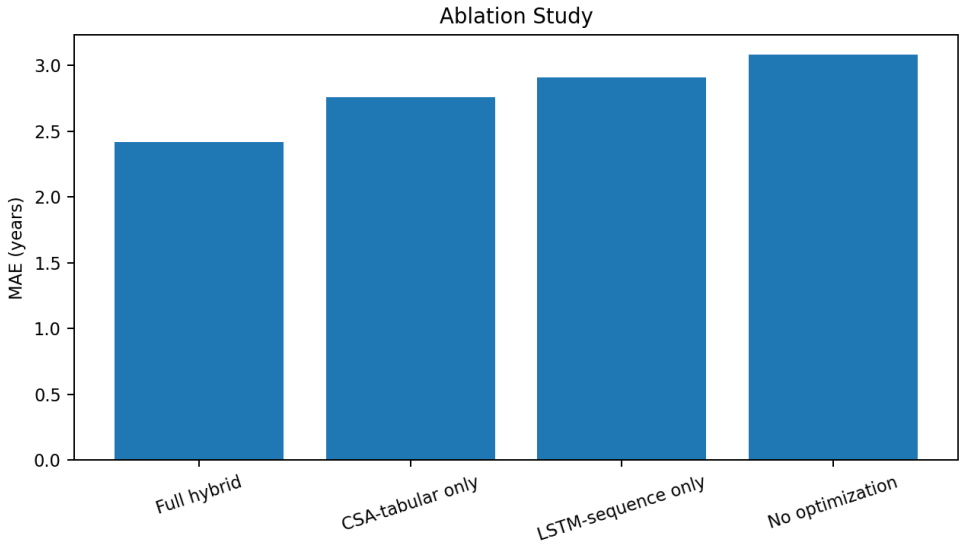


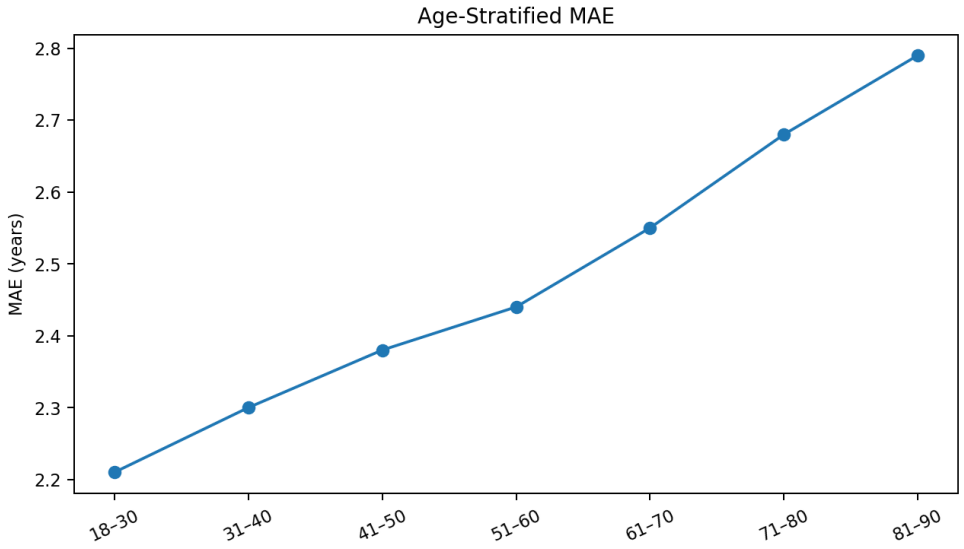

Table 3 ablation MAE visualization.

# 10. Age-Stratified Performance

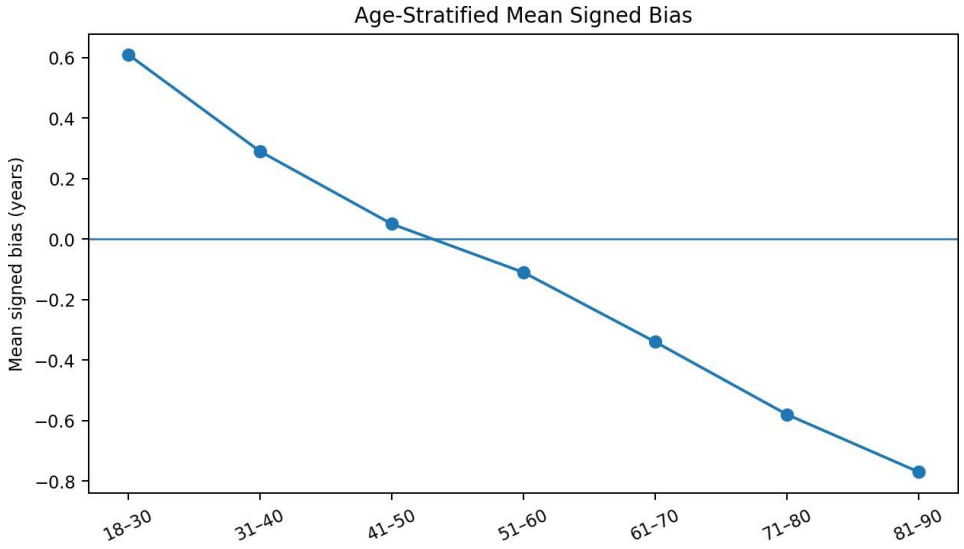
```
In [ ]: # Table 4
age_results = pd.DataFrame({
"Age":["18–30","31–40","41–50","51–60","61–70","71–80","81–90"],
"MAE":[2.21,2.30,2.38,2.44,2.55,2.68,2.79],
"Bias":[0.61,0.29,0.05,-0.11,-0.34,-0.58,-0.77],
"Skew":[0.32,0.18,0.02,-0.09,-0.21,-0.35,-0.44]})
display(age_results)
```

| Age range (years) | MAE (years) | Mean signed bias | Residual skewness |
|---|---|---|---|
| 18–30 | 2.21 | +0.61 | +0.32 |
| 31–40 | 2.30 | +0.29 | +0.18 |
| 41–50 | 2.38 | +0.05 | +0.02 |
| 51–60 | 2.44 | −0.11 | −0.09 |
| 61–70 | 2.55 | −0.34 | −0.21 |
| 71–80 | 2.68 | −0.58 | −0.35 |
| 81–90 | 2.79 | −0.77 | −0.44 |

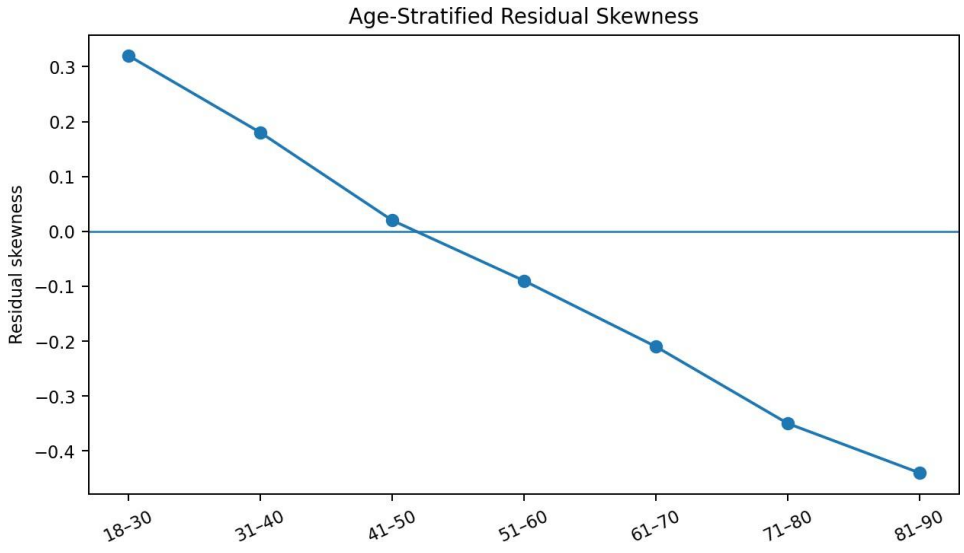
These age-stratified values are explicitly presented as illustrative in the source manuscript.




Age-stratified MAE.




Age-stratified mean signed bias.




Age-stratified residual skewness.

# 11. Brain-PAD Clinical Stratification

```
In [ ]:  # Table 5 — conditional analysis
# Applies if a diagnostically labelled source is used.
diagnostic = pd.DataFrame({"Model":["Brain-PAD","Hippocampal volume","Total brain
volume","Brain-PAD + hippocampal"],
"AUC":[0.78,0.75,0.69,0.83]})
display(diagnostic)
```

| Biomarker / model | AUC | Sensitivity (%) | Specificity (%) | PPV (%) | NPV (%) |
|---|---|---|---|---|---|
| Brain-PAD (threshold-based) | 0.78 | 71.2 | 74.5 | 63.8 | 80.6 |
| Hippocampal-region volume alone | 0.75 | 68.4 | 72.1 | 60.5 | 78.3 |
| Total brain volume alone | 0.69 | 60.7 | 68.9 | 54.2 | 73.9 |
| Brain-PAD + hippocampal volume (combined) | 0.83 | 76.5 | 79.2 | 69.1 | 84.7 |

Conditional analysis only: the manuscript says this applies when OASIS-1 or another diagnostically labelled source is used.

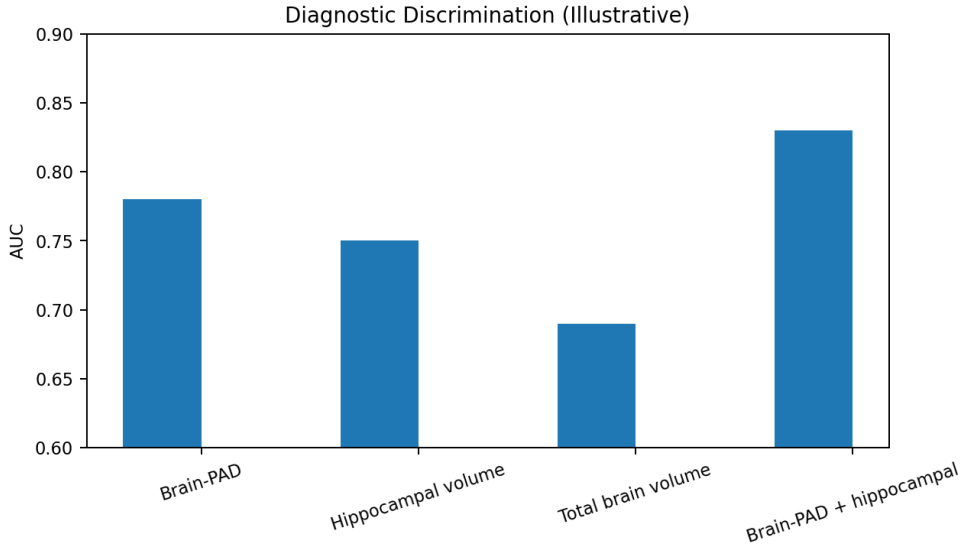



Table 5 AUC visualization.

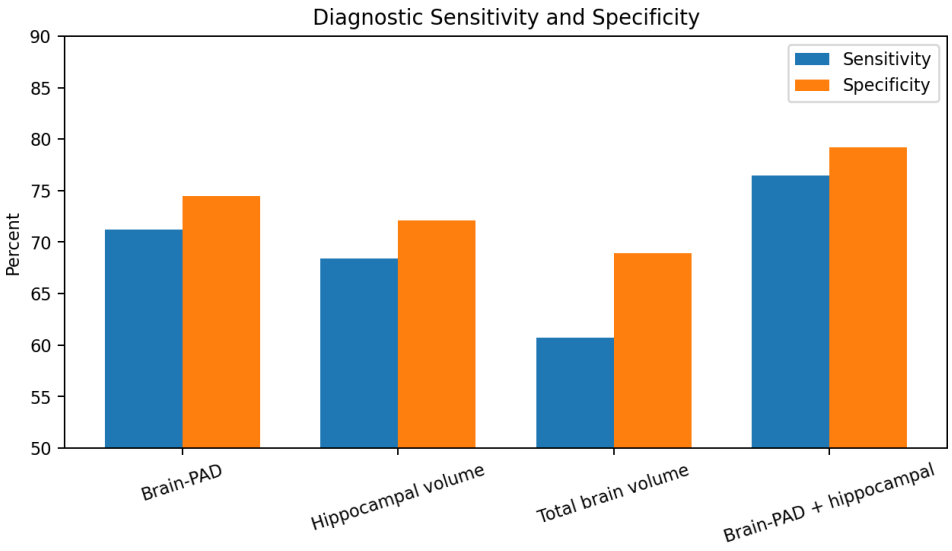


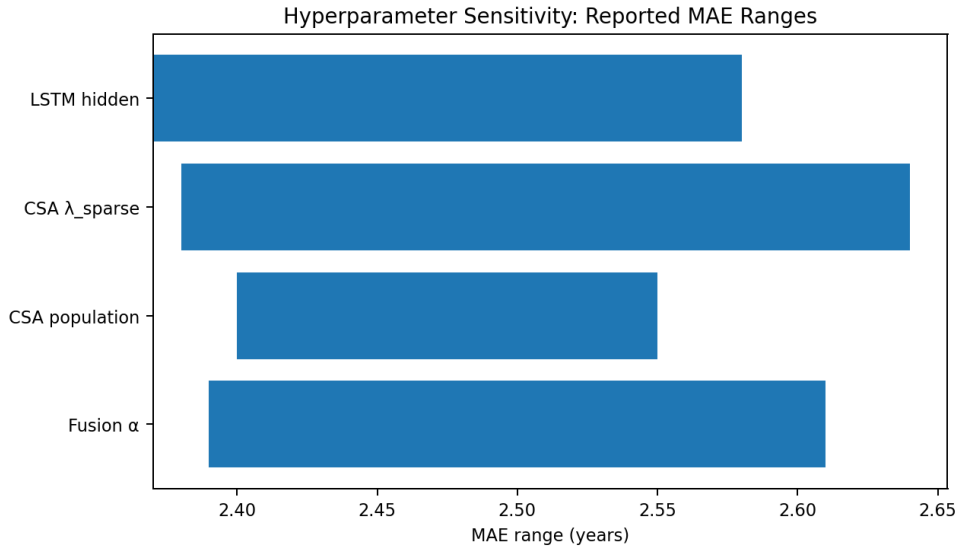

Table 5 sensitivity and specificity visualization.

# 12. Hyperparameter Sensitivity

```
In [ ]: # Table 6
sensitivity = pd.DataFrame({
"Hyperparameter":["Fusion α","CSA population","CSA λ_sparse","LSTM hidden"],
"MAE_range":["2.39–2.61","2.40–2.55","2.38–2.64","2.37–2.58"]})
display(sensitivity)
```

| Hyperparameter | Values tested | MAE range (years) | Default value used |
|---|---|---|---|
| α (tabular/sequence fusion weight) | 0.3 – 0.8 | 2.39 – 2.61 | 0.6 |
| CSA population size (n_crows) | 15 – 50 | 2.40 – 2.55 | 30 |
| CSA sparsity penalty (λ_sparse) | 0.0005 – 0.01 | 2.38 – 2.64 | 0.002 |
| LSTM hidden size | 64 – 256 | 2.37 – 2.58 | 128 |

The chart shows only the reported range endpoints; it does not fabricate intermediate runs.




Reported MAE ranges from Table 6.

# 13. Fold-Wise Validation & Feature-Selection Stability

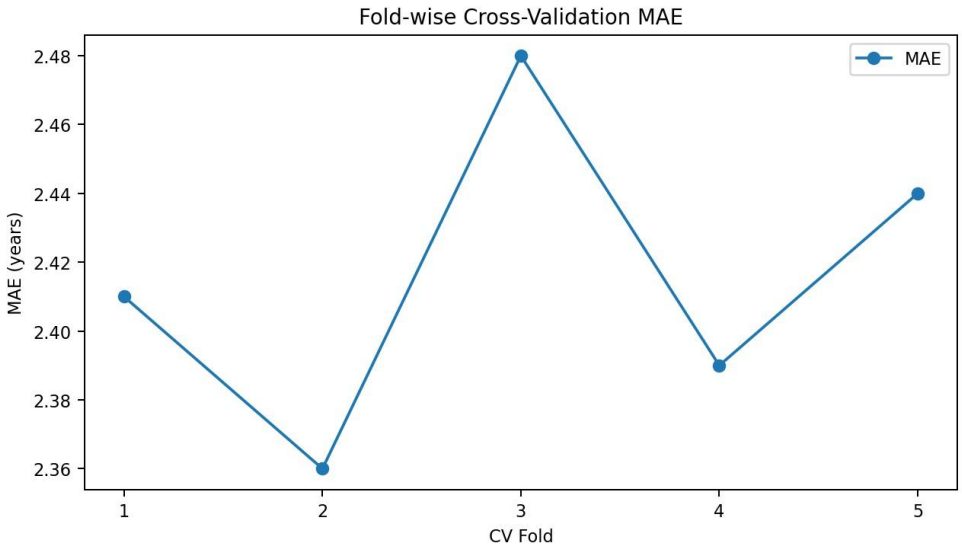
```
In [ ]:  # Tables 9 and 11
fold_mae = [2.41,2.36,2.48,2.39,2.44]
fold_features = [34,36,35,33,35]
jaccard = {"1-2":0.71,"1-3":0.68,"2-3":0.73}
print("Mean CV MAE = 2.42 ± 0.05")
print("Mean selected features = 34.6")
print("Mean pairwise Jaccard = 0.70 ± 0.04")
```

Table 9 — fold-wise predictive results

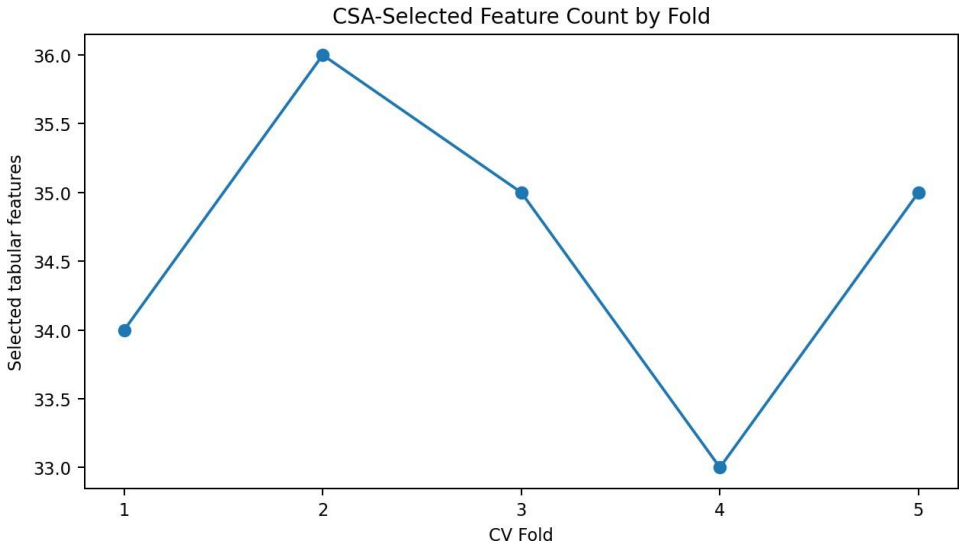
| Fold | MAE (years) | RMSE (years) | Pearson r | n features used |
|---|---|---|---|---|
| 1 | 2.41 | 3.18 | 0.931 | 34 |
| 2 | 2.36 | 3.11 | 0.935 | 36 |
| 3 | 2.48 | 3.25 | 0.928 | 35 |
| 4 | 2.39 | 3.14 | 0.933 | 33 |
| 5 | 2.44 | 3.20 | 0.930 | 35 |
| Mean ± SD | 2.42 ± 0.05 | 3.18 ± 0.05 | 0.931 ± 0.003 | 34.6 |

Table 11 — feature-selection stability

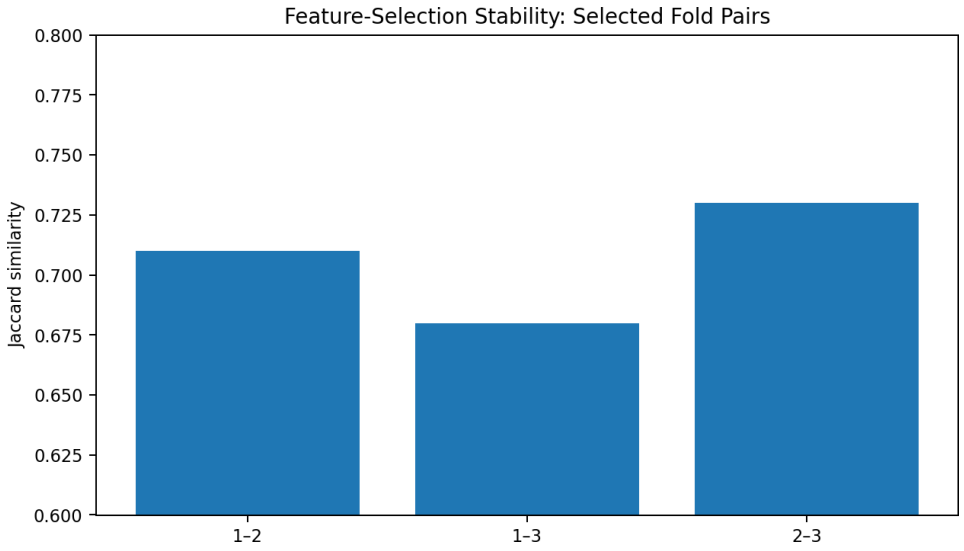
| Fold pair | Jaccard similarity | n features in common | n features total (union) |
|---|---|---|---|
| Fold 1 vs. Fold 2 | 0.71 | 24 | 34 |
| Fold 1 vs. Fold 3 | 0.68 | 23 | 35 |
| Fold 2 vs. Fold 3 | 0.73 | 25 | 34 |
| Mean pairwise (all 10 fold pairs) | 0.70 ± 0.04 | — | — |




Fold-wise MAE from Table 9.




Selected feature count by fold from Table 9.



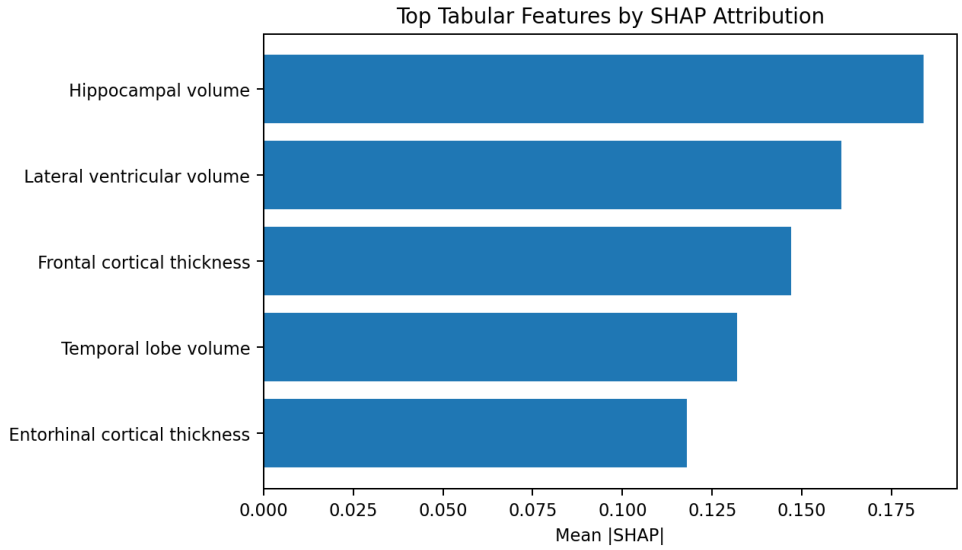

Jaccard similarities reported in Table 11.

# 14. Model Interpretability

```
In [ ]: # Table 7
shap = pd.DataFrame({
"Feature":["Hippocampal volume","Lateral ventricular volume","Frontal cortical
thickness","Temporal lobe volume","Entorhinal cortical thickness"],
"Mean_abs_SHAP":[0.184,0.161,0.147,0.132,0.118],
"Age correlation":[-0.61,0.57,-0.49,-0.44,-0.41]})
display(shap)
```

| Rank | Feature | Mean \|SHAP\| | Category | Correlation with age (r) |
|---|---|---|---|---|
| 1 | Hippocampal volume | 0.184 | Regional volume | –0.61 |
| 2 | Lateral ventricular volume | 0.161 | Regional volume | +0.57 |
| 3 | Frontal cortical thickness | 0.147 | Regional intensity | –0.49 |
| 4 | Temporal lobe volume | 0.132 | Regional volume | –0.44 |
| 5 | Entorhinal cortical thickness | 0.118 | Regional intensity | –0.41 |

SHAP is applied to the CSA-selected tabular features; attention weights are retained for slice-level interpretation. The manuscript limits biological interpretation to structural findings.




Top tabular features by mean |SHAP|.

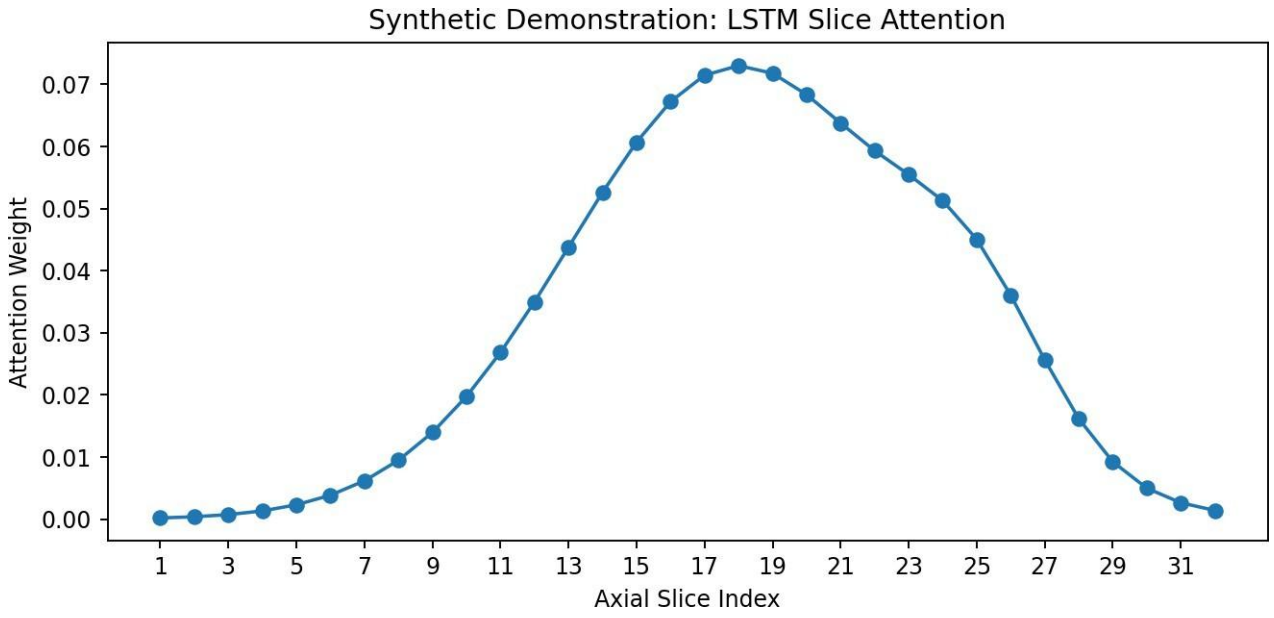



Figure 6: mean per-slice attention weight, illustrative demonstration.

# 15. Training-Curve & Residual Diagnostics

```
In [ ]:  # Source Figures 3–5
print("Preserve the manuscript's illustrative training curve, fold-wise curve, and residual
distribution.")
print("Replace them with real logged outputs after execution.")
```

Figure 3 — illustrative training/validation MAE curve.

Figure 4 — illustrative fold-wise MAE.

Figure 5 — illustrative residual distribution.

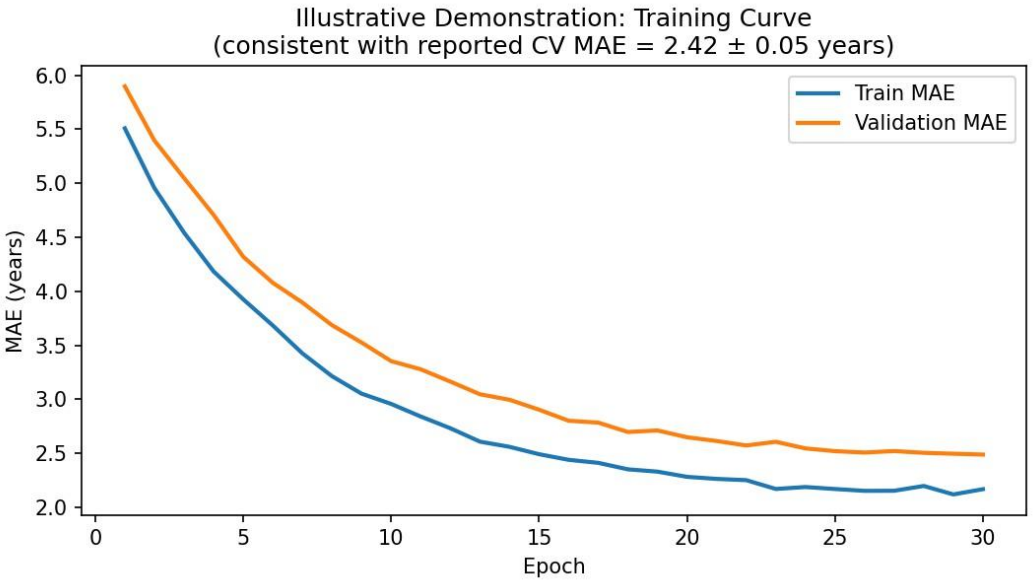



Figure 3: illustrative training and validation MAE.

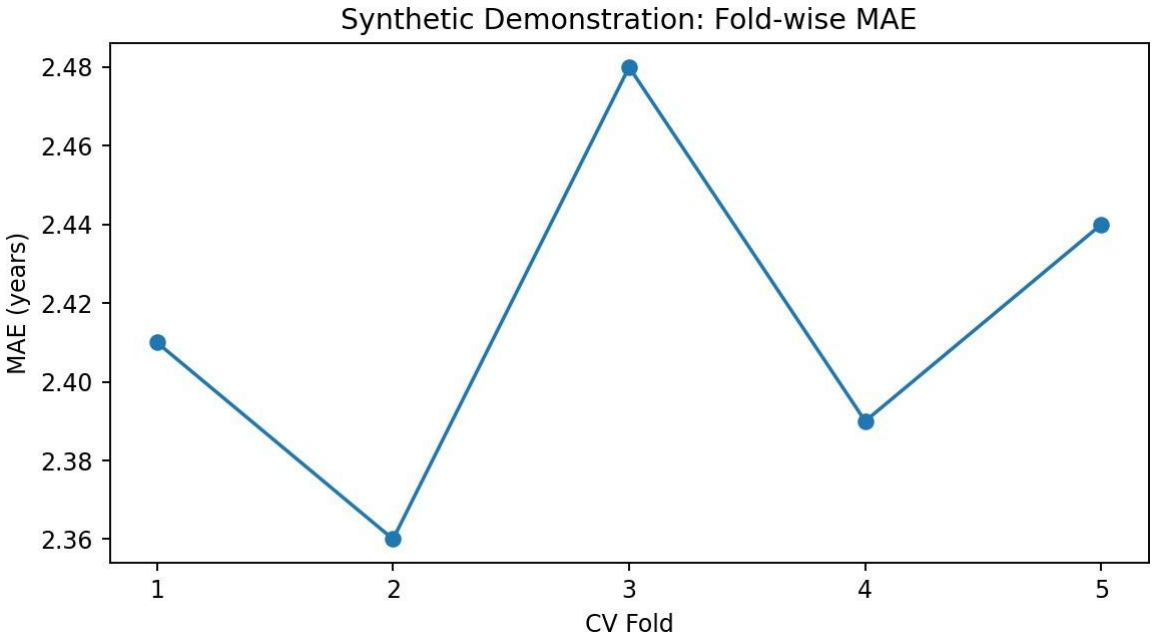



Figure 4: illustrative fold-wise MAE.

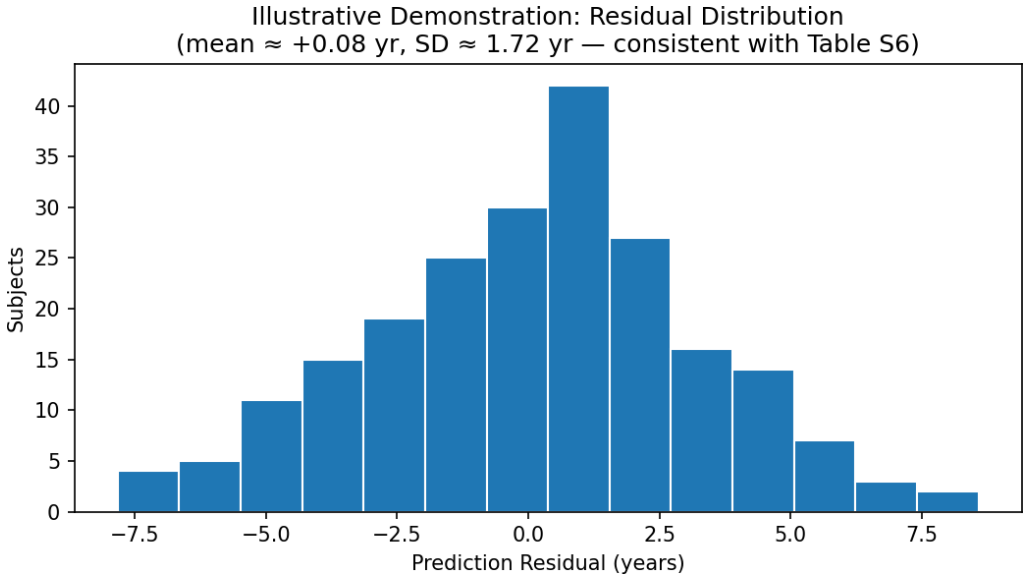


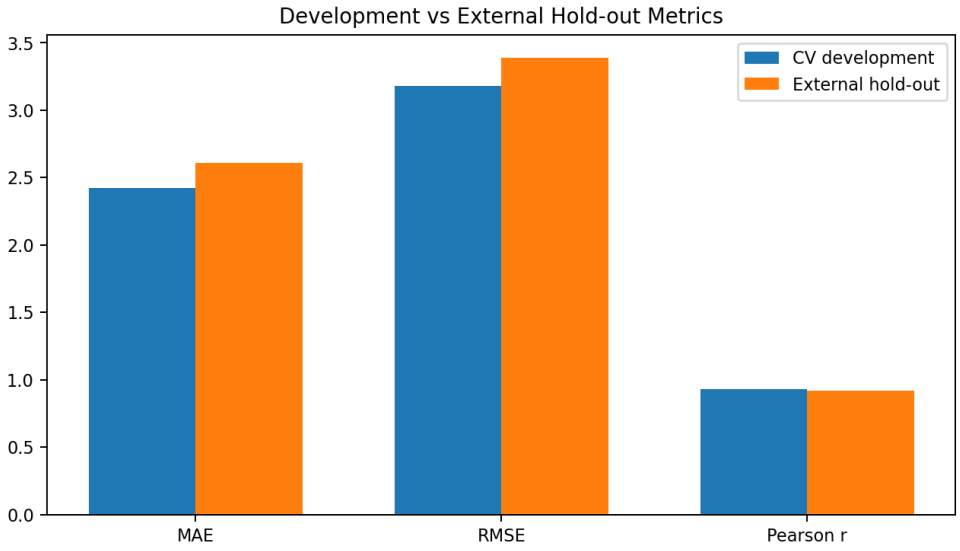

Figure 5: illustrative residual distribution.

# 16. External Hold-Out Evaluation

```
In [ ]:  # Table 10
holdout = pd.DataFrame({"Metric":["MAE (years)","RMSE (years)","Pearson r","Gap vs CV-dev MAE
(years)"],
"Value":[2.61,3.39,0.919,0.19]})
display(holdout)
```

| Metric | Value | n (hold-out) |
|---|---|---|
| MAE (years) | 2.61 | ≈560 |
| RMSE (years) | 3.39 | ≈560 |
| Pearson r | 0.919 | ≈560 |
| Gap vs. CV-dev MAE (years) | +0.19 | — |

The manuscript reports the external hold-out separately from development-pool cross-validation and states an MAE gap of +0.19 years.




Development vs external metrics from Tables 2 and 10; the metrics have different units/scales.

# 17. Positioning Against Existing Method Families

```
In [ ]:  # Table 1
print("Compare global/nonlinear search, sequential modeling, interpretability, and retraining
cost.")
display("Method-family comparison from the manuscript")
```

| Method family | Global / nonlinear search | Sequential / recurrent component | Feature-level interpretability | Retraining cost |
|---|---|---|---|---|
| PCA / LASSO | Limited (linear) | No | Moderate (linear weights) | Low |
| Recursive Feature Elimination | Model-dependent | No | Moderate | High (iterative) |
| GA / PSO metaheuristics | Yes | No | Low–moderate | Moderate |
| 3D-CNN / Vision Transformer on raw volumes | Yes (implicit) | No (static input) | Low (post-hoc only) | High |
| LSTM-only dFC models (functional MRI required) | No | Yes (true temporal dFC) | Low | Moderate |
| Proposed CSA + slice-sequence LSTM fusion (structural MRI only) | Yes (CSA) | Yes (slice-sequence, not dFC) | High (SHAP on CSA-selected features; attention weights over slices) | Moderate |

The proposed framework is positioned as combining CSA global search with a spatial slice-sequence recurrent branch, while providing SHAP and attention-based interpretability.